



SUJET STATAPP 333 - 2026-2027

Sujet proposé par : Belpaire Michael - michael.belpaire@symbioticsgroup.com
Organisme : Symbiotics Asset Management
Catégorie : Finance
Sujet :
Simplifying Corporate Default Risk Prediction Using Public Financial Data

Description du sujet :

Credit-risk analysis often relies on a large number of accounting, market and macroeconomic indicators, as well as increasingly complex statistical and machine-learning models. However, it is not always clear how much additional predictive value this complexity provides.

The objective of this project is to assess whether corporate financial distress and default risk can be predicted effectively using a small and interpretable set of publicly available financial and market indicators.

The study will focus primarily on publicly listed companies, initially in the United States, where long historical series of accounting and market data are available. Students will reproduce and compare well-documented approaches from the credit-risk literature, ranging from traditional accounting-based models to more recent market-based and machine-learning approaches.

The project will compare, in particular:

- traditional financial-ratio models inspired by Altman and Ohlson;
- logistic or hazard models inspired by Shumway and Campbell, Hilscher and Szilagyi;
- a machine-learning benchmark such as Random Forest, XGBoost or LightGBM;
- simplified versions of these models using progressively fewer variables.

The main research question will be:

- How many variables are really necessary to retain most of the predictive power of a more complex corporate default-risk model?

Students will evaluate model performance using out-of-sample and time-based validation, with particular attention to predictive discrimination, calibration and interpretability.

An additional objective will be to identify the main financial drivers of corporate distress and determine whether a parsimonious model based on approximately 5 to 10 variables can provide performance close to that of a larger model.

The project could ultimately lead to the construction of a simple and transparent Corporate Credit Risk Score.

Students should be comfortable with Python and standard statistical and machine-learning methods, particularly logistic regression, classification techniques and model evaluation.

Données du sujet :

The project will rely primarily on public or easily accessible historical data.

The main sources considered are:

- SEC EDGAR / SEC Company Facts API: financial statements of listed U.S. companies, including assets, liabilities, equity, revenues, net income, cash, debt, operating income and cash flows;
- Public market data: historical share prices, market capitalization, stock returns and equity volatility;
- Kenneth French Data Library: market and industry factors used to construct market-adjusted returns and risk indicators;
- Federal Reserve Economic Data (FRED): interest rates, corporate credit spreads and other macroeconomic indicators;
- Public bankruptcy and financial-distress information: used to construct the target variable indicating whether a company experiences financial distress or bankruptcy over a defined future horizon.

Modèles envisagés du sujet :

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- a machine-learning benchmark such as Random Forest, XGBoost or LightGBM;
- simplified versions of these models using progressively fewer variables.

Bibliographie du sujet :

Altman, E. I. (1968), Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy, Journal of Finance.

Ohlson, J. A. (1980), Financial Ratios and the Probabilistic Prediction of Bankruptcy, Journal of Accounting Research.

Shumway, T. (2001), Forecasting Bankruptcy More Accurately: A Simple Hazard Model, Journal of Business.

Campbell, J. Y., Hilscher, J. and Szilagyi, J. (2008), In Search of Distress Risk, Journal of Finance.

Duan, J.-C., Sun, J. and Wang, T. (2012), Multiperiod Corporate Default Prediction ? A Forward Intensity Approach, Journal of Econometrics.

Chan-Lau, J. A. et al. (2024), Monitoring Privately-held Firms' Default Risk in Real Time: A Signal-Knowledge Transfer Learning Model, IMF Working Paper.

U.S. Securities and Exchange Commission, EDGAR Application Programming Interfaces and Company Facts datasets.

Federal Reserve Bank of St. Louis, FRED Economic Data.

Kenneth R. French, Data Library

Notation du sujet :

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Commentaires :

Sujet adéquat